Canonical Shells and Residue-Cover Trees in a Conditional First-Descent Approach to the Collatz Problem

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Abstract

This paper studies delayed first descent for the accelerated odd Collatz map

$$T(n) = \frac{3n+1}{2^{v_2(3n+1)}}.$$

Building on the modular resistance-spine framework and certified spine-descent thresholds developed in earlier work, the positive odd integers are organized into disjoint canonical shells

$$S_m^{\circ} = \{n \in \mathbb{O}^+ : v_2(n+1) = m\}.$$

Within each nontrivial shell, certified elements descend exactly at the deterministic exit time m. The remaining non-certified component is studied through a dyadic gap condition, which forces any descent in that component to occur strictly after the exit time.

The delayed-descent problem for non-certified elements is then formulated through valuation cylinders and residue-cover trees. A closed cylinder gives a uniform cumulative descent certificate for all of its elements. Consequently, the dyadic gap condition together with closure of every non-certified residue-cover tree yields delayed descent for all non-certified canonical-shell elements. Combined with certified exact descent and immediate descent in the first canonical shell, this gives a conditional route to universal first-descent coverage. By the standard minimal-counterexample argument, such coverage implies Collatz convergence.

The framework isolates two structural tasks: proving the dyadic gap condition and proving closure of the non-certified residue-cover trees. Finite exact-integer computations support both components. The dyadic gap was verified for $2 \le m \le 100000$, and all $1,431,369$ tested non-certified canonical-shell cases up to $N = 10^7$ were resolved with delayed descent.

Keywords: Collatz conjecture; accelerated odd Collatz map; first-descent time; canonical shells; dyadic gap; continued fractions; valuation cylinders; residue-cover trees; non-certified delayed descent; exact integer computation.

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1 Introduction

The Collatz problem, also known as the $3x + 1$ problem, asks whether every positive integer eventually reaches 1 under the iteration

$$x \mapsto \begin{cases} x/2, & x \equiv 0 \pmod 2, \\ 3x+1, & x \equiv 1 \pmod 2. \end{cases}$$

Despite its elementary formulation, the problem remains one of the central open questions in elementary number theory and discrete dynamical systems. Classical work has studied the problem through stopping times, density phenomena, congruence structure, and computational verification [1, 2, 3, 4, 5]. Comprehensive accounts and modern overviews of the $3x+1$ problem are given in [6, 7, 8]. A major recent development is Tao's logarithmic-density result, which shows that almost all Collatz orbits attain almost bounded values [9]. Large-scale verification has also continued to advance, with recent computational work extending the verified range substantially [10].

A standard reduction is obtained by restricting the dynamics to positive odd integers and removing all powers of 2 after each odd step. This gives the accelerated odd Collatz map

$$T(n) = \frac{3n+1}{2^{v_2(3n+1)}}, \qquad n \in \mathbb{O}^+ = \{1, 3, 5, \ldots\}.$$

The map T records the odd-to-odd dynamics of the usual Collatz iteration and is a standard formulation of the accelerated $3x + 1$ dynamics [4, 8].

The present paper studies first descent below the initial value. For

$$n \in \mathbb{O}^+, \qquad n > 1,$$

the first-descent time is defined, whenever it exists, by

$$\tau(n) = \min\{r \geq 1 : T^r(n) < n\}.$$

A universal first-descent theorem for all odd

$$n > 1$$

would imply Collatz convergence by the standard minimal-counterexample argument. The purpose of this paper is to develop a conditional route toward such first-descent coverage through canonical shells, dyadic gaps, and residue-cover trees.

This work builds on two earlier components of the pre-descent program. The first introduced finite pre-descent resistance, relative peak growth, and modular resistance spines [11]. In that framework, the nested spine classes

$$S_m = \{n \in \mathbb{O}^+ : n \equiv -1 \pmod{2^m}\}$$

were shown to force deterministic resistance to early descent. In particular, if

$$n = 2^m q - 1,$$

then

$$T^j(n) = 3^j 2^{m-j} q - 1 \qquad (0 \leq j \leq m-1),$$

so descent below the initial value cannot occur during the first $m - 1$ accelerated odd iterates.

The second component developed exact descent inside these modular resistance spines [12]. For spine elements

$$n = 2^m q - 1,$$

the certified threshold

$$\kappa_m = \left\lceil \log_2 \left(\frac{3^m}{2^m - 1} \right) \right\rceil$$

and certified residue

$$q_m^* \equiv (3^m)^{-1} \pmod{2^{\kappa_m}}$$

were isolated. The certified class satisfies exact first descent:

$$q \equiv q_m^* \pmod{2^{\kappa_m}} \implies \tau(2^m q - 1) = m.$$

The shifted non-certified residues were also shown to have a structured first-exit valuation:

$$q \equiv q_m^* + a \pmod{2^{\kappa_m}}, \qquad 1 \leq a < 2^{\kappa_m},$$

implies

$$v_2(3^m q - 1) = v_2(a).$$

The present paper transfers this certified/non-certified framework from nested spines to a disjoint decomposition of all positive odd integers. The central organizing object is the canonical shell

$$S_m^\circ = \{n \in \mathbb{O}^+ : v_2(n+1) = m\}.$$

Equivalently,

$$S_m^\circ = S_m \setminus S_{m+1}.$$

Thus S_m is a nested modular spine, while S_m° is an exact canonical layer. The shells form the disjoint decomposition

$$\mathbb{O}^+ = \bigsqcup_{m \geq 1} S_m^\circ.$$

If

$$n \in S_m^\circ,$$

then

$$n = 2^m q - 1$$

for a unique odd integer q.

For $m \geq 2$, canonical-shell elements remain above their initial value during the first $m-1$ accelerated iterates. At the deterministic exit time m, the behavior is governed by

$$v_2(3^m q - 1).$$

This gives a certified/non-certified decomposition

$$S_m^\circ = \mathcal{D}_m^\circ \sqcup \mathcal{N}_m^\circ.$$

Certified elements descend exactly at time m. The non-certified component is analyzed through the dyadic gap condition

$$2^{K_m} - 3^m \geq 2^{K_m - m}, \qquad K_m = \lceil m \log_2 3 \rceil.$$

Under this condition, non-certified shell elements remain above their initial value at the deterministic exit time. Hence any descent in the non-certified component occurs strictly after the exit:

$$\tau(n) > m.$$

The dyadic gap condition is also studied as a Diophantine separation problem between powers of 2 and 3. A possible failure would imply an exponentially accurate upper rational approximation to

$$\log_2 3.$$

By Legendre's criterion for continued fractions, any failure level is forced to occur at the denominator of an upper continued-fraction convergent of

$$\log_2 3$$

[13, 14]. Exact-integer verification found no dyadic gap failure for

$$2 \leq m \leq 100000.$$

After the deterministic exit, delayed descent in the non-certified component is organized through valuation cylinders and residue-cover trees. A finite valuation pattern defines a congruence cylinder. Some cylinders carry a cumulative descent certificate strong enough to force every element in the cylinder below its initial value; these are called closed cylinders. If every branch of the non-certified residue-cover tree eventually reaches a closed cylinder, then the entire non-certified component has delayed descent.

The main conditional implication proved in this paper is

$$\text{Dyadic Gap} + \text{Non-Certified Tree Closure} \Longrightarrow \text{Full Non-Certified Delayed Coverage.}$$

Together with certified exact descent and immediate descent in the first canonical shell, this gives conditional universal first-descent coverage. By the standard minimal-counterexample argument, universal first-descent coverage implies Collatz convergence.

The framework isolates two structural tasks:

the dyadic gap condition

and

closure of every non-certified residue-cover tree.

Establishing these two tasks would complete the first-descent route developed here.

2 Preliminaries

Let

$$\mathbb{O}^+ = \{1, 3, 5, \dots\}$$

denote the set of positive odd integers. For a positive integer x, let

$$v_2(x)$$

denote the exponent of the highest power of 2 dividing x. Thus

$$2^{v_2(x)} \mid x \qquad \text{and} \qquad 2^{v_2(x)+1} \nmid x.$$

The accelerated odd Collatz map is

$$T : \mathbb{O}^+ \to \mathbb{O}^+, \qquad T(n) = \frac{3n+1}{2^{v_2(3n+1)}}.$$

Since $3n + 1$ is even for every odd n, the valuation $v_2(3n + 1)$ is at least 1. Dividing by the full power of 2 leaves an odd integer, so T maps $\mathbb{O}^+$ into itself.

For $r \geq 0$, the iterates of T are denoted by

$$T^0(n) = n, \qquad T^{r+1}(n) = T(T^r(n)).$$

Definition 2.1 (First-Descent Time)**.** Let

$$n \in \mathbb{O}^+, \qquad n > 1.$$

If there exists an integer $r \geq 1$ such that

$$T^r(n) < n,$$

then the first-descent time of n is

$$\tau(n) = \min\{r \geq 1 : T^r(n) < n\}.$$

If no such integer exists, then $\tau(n)$ is left undefined.

Definition 2.2 (Canonical Shells)**.** For $m \geq 1$, the m-th canonical shell is

$$S_m^\circ = \{n \in \mathbb{O}^+ : v_2(n+1) = m\}.$$

The next lemma records that the canonical shells form a disjoint and exhaustive decomposition of the positive odd integers.

Lemma 2.3 (Canonical Shell Decomposition)**.** *The positive odd integers decompose as*

$$\mathbb{O}^+ = \bigsqcup_{m \geq 1} S_m^\circ.$$

Moreover,

$$n \in S_m^\circ$$

if and only if there exists a unique odd integer $q \geq 1$ such that

$$n = 2^m q - 1.$$

Proof. Let $n \in \mathbb{O}^+$. Since n is odd, $n+1$ is even. Hence

$$m = v_2(n+1)$$

is a well-defined integer with $m \geq 1$. Therefore $n \in S_m^\circ$. The value of m is unique, so the shells are disjoint and exhaustive.

Now suppose $n \in S_m^\circ$. Then

$$v_2(n+1) = m,$$

so

$$n + 1 = 2^m q$$

for a unique integer $q \geq 1$. Since the valuation is exactly m, the integer q is odd. Hence

$$n = 2^m q - 1.$$

Conversely, if

$$n = 2^m q - 1$$

with $q \geq 1$ odd, then

$$n + 1 = 2^m q.$$

Since q is odd,

$$v_2(n+1) = m.$$

Thus $n \in S_m^\circ$. □

3 Deterministic Persistence in Canonical Shells

The first structural feature of a canonical shell is deterministic persistence. This identity was used in the modular resistance-spine setting in [11]. Here it is used in its canonical-shell form.

Lemma 3.1 (Persistence Identity)**.** *Let*

$$m \geq 1$$

and let

$$n = 2^m q - 1$$

with $q \geq 1$ odd. Then, for every

$$0 \leq j \leq m - 1,$$

one has

$$T^j(n) = 3^j 2^{m-j} q - 1.$$

Proof. The case $j = 0$ is immediate. Suppose the formula holds for some

$$0 \leq j < m - 1.$$

Then

$$T^j(n) = 3^j 2^{m-j} q - 1.$$

Hence

$$3T^j(n) + 1 = 3^{j+1} 2^{m-j} q - 2 = 2\left(3^{j+1} 2^{m-j-1} q - 1\right).$$

Since $m - j - 1 \geq 1$, the integer

$$3^{j+1} 2^{m-j-1} q$$

is even, so

$$3^{j+1} 2^{m-j-1} q - 1$$

is odd. Therefore

$$v_2(3T^j(n) + 1) = 1,$$

and so

$$T^{j+1}(n) = \frac{3T^j(n) + 1}{2} = 3^{j+1} 2^{m-j-1} q - 1.$$

This completes the induction. □

Corollary 3.2 (No Early Descent Before the Exit)**.** *Let*

$$m \geq 2$$

and let

$$n = 2^m q - 1 \in S_m^\circ$$

with $q \geq 1$ odd. Then

$$T^j(n) > n \qquad (1 \leq j \leq m - 1).$$

Consequently, descent below the initial value cannot occur during the first $m - 1$ accelerated odd iterates.

Proof. By Theorem 3.1,

$$T^j(n) = 3^j 2^{m-j} q - 1.$$

Therefore

$$T^j(n) - n = 3^j 2^{m-j} q - 1 - (2^m q - 1) = 2^{m-j} q (3^j - 2^j).$$

For

$$1 \leq j \leq m - 1,$$

one has

$$3^j > 2^j.$$

Since

$$2^{m-j}q > 0,$$

it follows that

$$T^j(n) > n.$$

□

4 First Exit and Certified Shell Decomposition

Let

$$n = 2^m q - 1 \in S_m^\circ$$

with $m \geq 2$. By the canonical shell decomposition, q is a positive odd integer. The persistence identity gives

$$T^{m-1}(n) = 3^{m-1}2q - 1.$$

Therefore

$$3T^{m-1}(n) + 1 = 2(3^m q - 1).$$

Thus the m-th accelerated iterate is governed by the valuation of

$$3^m q - 1.$$

Lemma 4.1 (First-Exit Formula)**.** *Let*

$$m \geq 2$$

and let

$$n = 2^m q - 1 \in S_m^\circ.$$

Set

$$s = v_2(3^m q - 1).$$

Then

$$T^m(n) = \frac{3^m q - 1}{2^s}.$$

Proof. By Theorem 3.1,

$$T^{m-1}(n) = 3^{m-1}2q - 1.$$

Hence

$$3T^{m-1}(n) + 1 = 2(3^m q - 1).$$

Since

$$s = v_2(3^m q - 1),$$

one has

$$v_2(3T^{m-1}(n) + 1) = s + 1.$$

Therefore

$$T^m(n) = \frac{3T^{m-1}(n) + 1}{2^{s+1}} = \frac{3^m q - 1}{2^s}.$$

□

The following threshold and certified residue are the canonical-shell versions of the exact-descent threshold isolated in the spine setting in [12].

Definition 4.2 (Certified Threshold)**.** For $m \geq 2$, define

$$\kappa_m = \left\lceil \log_2 \left(\frac{3^m}{2^m - 1} \right) \right\rceil.$$

Equivalently, κ_m is the least integer $k \geq 1$ such that

$$2^k (2^m - 1) \geq 3^m.$$

Definition 4.3 (Certified Residue)**.** For $m \geq 2$, let

$$q_m^*$$

denote the unique residue class modulo 2^{κ_m} satisfying

$$3^m q_m^* \equiv 1 \pmod{2^{\kappa_m}}.$$

Equivalently,

$$q_m^* \equiv (3^m)^{-1} \pmod{2^{\kappa_m}}.$$

When a representative is needed, the least positive representative is used.

Definition 4.4 (Certified and Non-Certified Canonical Components)**.** Let $m \geq 2$. The certified component of S_m° is

$$\mathcal{D}_m^\circ = \left\{ 2^m q - 1 \in S_m^\circ : q \equiv q_m^* \pmod{2^{\kappa_m}} \right\}.$$

The non-certified component is

$$\mathcal{N}_m^\circ = S_m^\circ \setminus \mathcal{D}_m^\circ.$$

Thus

$$S_m^\circ = \mathcal{D}_m^\circ \sqcup \mathcal{N}_m^\circ.$$

Theorem 4.5 (Certified Exact Descent)**.** *Let*

$$m \geq 2$$

and let

$$n = 2^m q - 1 \in \mathcal{D}_m^\circ.$$

Then

$$T^m(n) < n.$$

Moreover,

$$T^j(n) > n \qquad (1 \leq j \leq m - 1).$$

Consequently,

$$\tau(n) = m.$$

Proof. Since

$$n \in \mathcal{D}_m^\circ,$$

one has

$$q \equiv q_m^* \pmod{2^{\kappa_m}}.$$

Hence

$$3^m q \equiv 1 \pmod{2^{\kappa_m}},$$

so

$$s = \nu_2(3^m q - 1) \geq \kappa_m.$$

By the first-exit formula,

$$T^m(n) = \frac{3^m q - 1}{2^s}.$$

Since

$$s \geq \kappa_m,$$

the definition of κ_m gives

$$2^s(2^m - 1) \geq 3^m.$$

Multiplying by $q \geq 1$, one obtains

$$2^s q(2^m - 1) \geq 3^m q.$$

Also,

$$2^s(2^m q - 1) = 2^s q(2^m - 1) + 2^s(q - 1).$$

Since $q \geq 1$, the second term is nonnegative. Therefore

$$2^s(2^m q - 1) \geq 3^m q > 3^m q - 1.$$

Dividing by 2^s gives

$$2^m q - 1 > \frac{3^m q - 1}{2^s}.$$

Thus

$$T^m(n) < n.$$

By Theorem 3.2,

$$T^j(n) > n \qquad (1 \leq j \leq m - 1).$$

Hence the first descent occurs exactly at time m:

$$\tau(n) = m.$$

□

5 The Dyadic Gap and the Non-Certified First-Exit Obstruction

Certified elements descend exactly at the deterministic exit time. The non-certified complement is governed by a different first-exit mechanism. Under a dyadic gap condition, non-certified shell elements remain above their initial value at the deterministic exit time. Thus any descent in the non-certified component is necessarily delayed.

The key point is to relate the certified threshold

$$\kappa_m$$

to a separation between powers of 2 and powers of 3.

Definition 5.1 (Dyadic Gap Condition)**.** For $m \geq 2$, set

$$K_m = \lceil m \log_2 3 \rceil .$$

The dyadic gap condition holds at level m if

$$2^{K_m} - 3^m \geq 2^{K_m - m}.$$

It holds globally if this inequality holds for every $m \geq 2$.

Lemma 5.2 (Threshold Identity Equivalent to the Dyadic Gap)**.** *Let*

$$K_m = \lceil m \log_2 3 \rceil$$

and

$$F_m = \left\lfloor m \log_2 \left(\frac{3}{2} \right) \right\rfloor .$$

For $m \geq 2$, the identity

$$\kappa_m = F_m + 1$$

holds if and only if

$$2^{K_m} - 3^m \geq 2^{K_m - m}.$$

Proof. By definition,

$$\kappa_m = \min\{k \geq 1 : 2^k (2^m - 1) \geq 3^m\}.$$

The number

$$\log_2 \left(\frac{3}{2} \right)$$

is irrational. Indeed, if

$$\log_2 \left(\frac{3}{2} \right) = \frac{a}{b}$$

with positive integers a, b, then

$$2^a = \left(\frac{3}{2} \right)^b ,$$

or equivalently

$$2^{a+b} = 3^b,$$

which contradicts unique prime factorization.

Hence

$$m \log_2 \left(\frac{3}{2} \right)$$

is not an integer, and therefore

$$F_m < m \log_2 \left(\frac{3}{2} \right) .$$

It follows that

$$2^{F_m} < \left(\frac{3}{2} \right)^m .$$

Multiplying by $2^m - 1 < 2^m$, one obtains

$$2^{F_m} (2^m - 1) < \left(\frac{3}{2} \right)^m (2^m - 1) < 3^m.$$

Thus F_m is too small to satisfy the defining inequality for κ_m, so

$$\kappa_m \geq F_m + 1.$$

The condition that $F_m + 1$ satisfies the threshold inequality is

$$2^{F_m+1}(2^m - 1) \geq 3^m.$$

This is equivalent to

$$2^{m+F_m+1} - 3^m \geq 2^{F_m+1}.$$

Since

$$m \log_2 3 = m + m \log_2 \left(\frac{3}{2}\right),$$

and

$$m \log_2 \left(\frac{3}{2}\right)$$

is not an integer, one has

$$K_m = \lceil m \log_2 3 \rceil = m + F_m + 1.$$

Substituting this into the previous inequality gives

$$2^{K_m} - 3^m \geq 2^{K_m - m}.$$

Therefore $F_m + 1$ is sufficient for the certified threshold exactly when the dyadic gap holds. Since F_m is always insufficient, this is equivalent to

$$\kappa_m = F_m + 1.$$

□

Proposition 5.3 (Dyadic Gap and Threshold Identity)**.** *Assume the dyadic gap condition at level m. Then*

$$\kappa_m = \left\lfloor m \log_2 \left(\frac{3}{2}\right) \right\rfloor + 1.$$

Proof. This follows directly from Theorem 5.2. □

Remark 5.4 (Meaning of the Dyadic Gap Condition)**.** The dyadic gap condition asserts that 3^m is separated from the next larger power of 2,

$$2^{K_m}, \qquad K_m = \lceil m \log_2 3 \rceil,$$

by at least

$$2^{K_m - m}.$$

Equivalently,

$$2^{K_m} - 3^m \geq 2^{K_m - m}.$$

Thus the condition is a Diophantine separation statement between powers of 2 and powers of 3. In this paper it is used as a structural hypothesis for the delayed-coverage theorem, and its possible failure levels are later constrained through continued-fraction approximation of

$$\log_2 3.$$

Lemma 5.5 (First-Exit Comparison)**.** *Let*

$$m \geq 2$$

and let

$$n = 2^m q - 1 \in S_m^\circ.$$

Set

$$s = v_2(3^m q - 1).$$

Then

$$T^m(n) > n$$

if and only if

$$q(3^m - 2^{m+s}) + 2^s - 1 > 0.$$

Proof. By Theorem 4.1,

$$T^m(n) = \frac{3^m q - 1}{2^s}.$$

Thus

$$T^m(n) > n$$

is equivalent to

$$\frac{3^m q - 1}{2^s} > 2^m q - 1.$$

Multiplying by 2^s and rearranging gives

$$3^m q - 1 > 2^{m+s} q - 2^s$$

if and only if

$$q(3^m - 2^{m+s}) + 2^s - 1 > 0.$$

The steps are reversible, so the equivalence follows. □

Theorem 5.6 (First-Exit Obstruction in Non-Certified Classes)**.** *Assume the dyadic gap condition at level m. Let*

$$n = 2^m q - 1 \in \mathcal{N}_m^\circ.$$

Then

$$T^m(n) > n.$$

Thus each non-certified shell element remains above its initial value at the deterministic exit time m.

Proof. Set

$$s = v_2(3^m q - 1).$$

Since

$$n \in \mathcal{N}_m^\circ,$$

the certified congruence fails:

$$q \not\equiv q_m^* \pmod{2^{\kappa_m}}.$$

Therefore

$$3^m q \not\equiv 1 \pmod{2^{\kappa_m}},$$

and hence

$$s < \kappa_m.$$

By the dyadic gap condition and Theorem 5.3,

$$\kappa_m = \left\lfloor m \log_2\left(\frac{3}{2}\right)\right\rfloor + 1.$$

Therefore

$$s \leq \left\lfloor m \log_2\left(\frac{3}{2}\right)\right\rfloor.$$

Since

$$m \log_2\left(\frac{3}{2}\right)$$

is not an integer, this gives

$$s < m \log_2\left(\frac{3}{2}\right).$$

Hence

$$2^s < \left(\frac{3}{2}\right)^m,$$

and multiplying by 2^m yields

$$2^{m+s} < 3^m.$$

Thus

$$3^m - 2^{m+s} > 0.$$

Because $q \geq 1$,

$$q(3^m - 2^{m+s}) > 0.$$

Also, since q is odd, $3^m q - 1$ is even, so

$$s \geq 1$$

and hence

$$2^s - 1 > 0.$$

Therefore

$$q(3^m - 2^{m+s}) + 2^s - 1 > 0.$$

By Theorem 5.5, this is equivalent to

$$T^m(n) > n.$$

□

Corollary 5.7 (Non-Certified Descent Must Be Delayed)**.** *Assume the dyadic gap condition at level m. Let*

$$n \in \mathcal{N}_m^\circ.$$

If there exists $r \geq 1$ such that

$$T^r(n) < n,$$

then

$$r > m.$$

Proof. Since

$$n \in \mathcal{N}_m^\circ \subseteq S_m^\circ,$$

Theorem 3.2 gives

$$T^j(n) > n \qquad (1 \le j \le m-1).$$

By Theorem 5.6,

$$T^m(n) > n.$$

Thus any descent time must occur strictly after the deterministic exit time:

$$r > m.$$

□

6 Diophantine Reduction of the Dyadic Gap

The dyadic gap condition can be reformulated as a Diophantine approximation condition for

$$\log_2 3.$$

This section shows that a possible dyadic-gap obstruction is forced onto a sparse set of continued-fraction denominators.

Let

$$\alpha = \log_2 3.$$

For $m \ge 2$, set

$$K_m = \lceil m\alpha \rceil.$$

Since α is irrational,

$$K_m - 1 < m\alpha < K_m.$$

Hence

$$2^{K_m - 1} < 3^m < 2^{K_m},$$

so 2^{K_m} is the next power of 2 above 3^m.

Define

$$\delta_m = K_m - m\alpha.$$

Then

$$0 < \delta_m < 1.$$

Lemma 6.1 (Fractional-Part Form of the Dyadic Gap)**.** *The dyadic gap condition*

$$2^{K_m} - 3^m \ge 2^{K_m - m}$$

is equivalent to

$$\delta_m \ge -\log_2(1 - 2^{-m}).$$

Proof. The dyadic gap condition is equivalent to

$$1 - \frac{3^m}{2^{K_m}} \geq 2^{-m}.$$

Since

$$3^m = 2^{m\alpha},$$

one has

$$\frac{3^m}{2^{K_m}} = 2^{m\alpha - K_m} = 2^{-\delta_m}.$$

Thus the condition becomes

$$1 - 2^{-\delta_m} \geq 2^{-m}.$$

Equivalently,

$$2^{-\delta_m} \leq 1 - 2^{-m}.$$

Taking $-\log_2$ gives

$$\delta_m \geq -\log_2(1 - 2^{-m}).$$

□

Lemma 6.2 (Failure Implies Exponential Rational Approximation)**.** *If the dyadic gap condition fails at level $m \geq 2$, then*

$$0 < K_m - m\log_2 3 < -\log_2(1 - 2^{-m}).$$

Consequently,

$$0 < \frac{K_m}{m} - \log_2 3 < \frac{-\log_2(1 - 2^{-m})}{m}.$$

In particular,

$$\left|\log_2 3 - \frac{K_m}{m}\right| < \frac{2^{1-m}}{m\log 2}.$$

Proof. By Theorem 6.1, dyadic gap failure gives

$$\delta_m < -\log_2(1 - 2^{-m}).$$

Since

$$\delta_m = K_m - m\log_2 3$$

and $\log_2 3$ is irrational, one has

$$\delta_m > 0.$$

Therefore

$$0 < K_m - m\log_2 3 < -\log_2(1 - 2^{-m}).$$

Dividing by m gives

$$0 < \frac{K_m}{m} - \log_2 3 < \frac{-\log_2(1 - 2^{-m})}{m}.$$

For $m \geq 2$,

$$0 < 2^{-m} \leq \frac{1}{4}.$$

Using

$$-\log(1 - x) < 2x \qquad (0 < x \leq 1/4),$$

with $x = 2^{-m}$, one obtains

$$-\log(1 - 2^{-m}) < 2^{1-m}.$$

Dividing by $\log 2$ gives

$$-\log_2(1 - 2^{-m}) < \frac{2^{1-m}}{\log 2}.$$

Thus

$$\left|\log_2 3 - \frac{K_m}{m}\right| < \frac{2^{1-m}}{m \log 2}.$$

□

Lemma 6.3 (Failure Approximation Beats Legendre Bound)**.** *For every*

$$m \geq 5,$$

one has

$$\frac{2^{1-m}}{m \log 2} < \frac{1}{2m^2}.$$

Proof. The desired inequality is equivalent to

$$m2^{2-m} < \log 2.$$

For $m = 5$,

$$5 \cdot 2^{-3} = \frac{5}{8} < \log 2.$$

Moreover, the sequence

$$m2^{2-m}$$

is decreasing for $m \geq 3$. Therefore

$$m2^{2-m} < \log 2$$

for every $m \geq 5$, which proves the claim. □

Proposition 6.4 (Continued-Fraction Reduction of Dyadic Gap Failure)**.** *If the dyadic gap condition fails at some level*

$$m \geq 5,$$

then

$$\frac{K_m}{m}, \qquad K_m = \lceil m \log_2 3 \rceil,$$

is an upper continued-fraction convergent of

$$\log_2 3.$$

In particular, m is the denominator of an upper continued-fraction convergent of

$$\log_2 3.$$

Proof. By Theorem 6.2, dyadic gap failure at level m gives

$$\left|\log_2 3 - \frac{K_m}{m}\right| < \frac{2^{1-m}}{m \log 2}.$$

For $m \geq 5$, Theorem 6.3 gives

$$\frac{2^{1-m}}{m \log 2} < \frac{1}{2m^2}.$$

Hence

$$\left|\log_2 3 - \frac{K_m}{m}\right| < \frac{1}{2m^2}.$$

By Legendre's criterion for continued fractions, the rational number

$$\frac{K_m}{m}$$

is a continued-fraction convergent of

$$\log_2 3$$

[13, 14]. Since

$$K_m = \lceil m \log_2 3 \rceil$$

and $\log_2 3$ is irrational, one has

$$\frac{K_m}{m} > \log_2 3.$$

Thus the convergent is an upper convergent. □

Lemma 6.5 (Small Levels)**.** *The dyadic gap condition holds for*

$$m = 2, 3, 4.$$

Proof. For $m = 2$,

$$K_2 = \lceil 2 \log_2 3 \rceil = 4,$$

and

$$2^{K_2} - 3^2 = 16 - 9 = 7, \qquad 2^{K_2 - 2} = 4.$$

Thus

$$7 \geq 4.$$

For $m = 3$,

$$K_3 = \lceil 3 \log_2 3 \rceil = 5,$$

and

$$2^{K_3} - 3^3 = 32 - 27 = 5, \qquad 2^{K_3 - 3} = 4.$$

Thus

$$5 \geq 4.$$

For $m = 4$,

$$K_4 = \lceil 4 \log_2 3 \rceil = 7,$$

and

$$2^{K_4} - 3^4 = 128 - 81 = 47, \qquad 2^{K_4 - 4} = 8.$$

Thus

$$47 \geq 8.$$

□

Corollary 6.6 (Sparse Obstruction Set for the Dyadic Gap)**.** *If the dyadic gap condition fails at any level*

$$m \geq 2,$$

then m is the denominator of an upper continued-fraction convergent of

$$\log_2 3.$$

Proof. By Theorem 6.5, the levels

$$m = 2, 3, 4$$

satisfy the dyadic gap condition. Hence any failure level satisfies

$$m \geq 5.$$

The conclusion follows from Theorem 6.4. □

Remark 6.7 (Scope of the Reduction)**.** The preceding result reduces the dyadic-gap obstruction to a sparse continued-fraction obstruction. More precisely, any failure level must occur among the denominators of upper continued-fraction convergents of

$$\log_2 3.$$

7 Cumulative Valuation Patterns

The cumulative affine form of the accelerated odd Collatz iteration is recorded in this section. This type of affine expansion is standard in the study of accelerated Collatz dynamics [4, 8], but the notation needed for the residue-cylinder framework is developed here explicitly.

Let

$$n \in \mathbb{O}^+.$$

For each $i \geq 0$, the i-th accelerated valuation is defined by

$$a_i(n) = v_2(3T^i(n) + 1).$$

Thus

$$T^{i+1}(n) = \frac{3T^i(n) + 1}{2^{a_i(n)}}.$$

Definition 7.1 (Finite Valuation Pattern)**.** For $r \geq 1$, a finite valuation pattern of length r is a tuple

$$\mathbf{a} = (a_0, \ldots, a_{r-1})$$

such that

$$a_i \geq 1$$

for every

$$0 \leq i \leq r - 1.$$

Definition 7.2 (Cumulative Valuation and Affine Offset)**.** Let

$$\mathbf{a} = (a_0, \ldots, a_{r-1})$$

be a valuation pattern of length r. The cumulative valuation sums are defined by

$$A_0(\mathbf{a}) = 0$$

and, for $1 \le j \le r$,

$$A_j(\mathbf{a}) = \sum_{i=0}^{j-1} a_i.$$

The affine offsets are defined recursively by

$$B_0(\mathbf{a}) = 0$$

and

$$B_{j+1}(\mathbf{a}) = 3B_j(\mathbf{a}) + 2^{A_j(\mathbf{a})} \qquad (0 \le j \le r-1).$$

Theorem 7.3 (Cumulative Affine Formula)**.** *Let*

$$n \in \mathbb{O}^+$$

and suppose that the first r accelerated valuations of n are given by

$$\mathbf{a} = (a_0(n), \ldots, a_{r-1}(n)).$$

Then

$$T^r(n) = \frac{3^r n + B_r(\mathbf{a})}{2^{A_r(\mathbf{a})}}.$$

Proof. The formula is proved by induction on r.

For $r = 0$, one has

$$T^0(n) = n.$$

Also,

$$A_0 = 0, \qquad B_0 = 0.$$

Hence

$$\frac{3^0 n + B_0}{2^{A_0}} = n.$$

Assume that the formula holds for some $r \ge 0$:

$$T^r(n) = \frac{3^r n + B_r}{2^{A_r}}.$$

Then

$$T^{r+1}(n) = \frac{3T^r(n) + 1}{2^{a_r}}.$$

Substitution of the induction hypothesis gives

$$T^{r+1}(n) = \frac{3\,(3^r n + B_r) + 2^{A_r}}{2^{A_r + a_r}}.$$

Since

$$A_{r+1} = A_r + a_r,$$

this becomes

$$T^{r+1}(n) = \frac{3^{r+1} n + 3B_r + 2^{A_r}}{2^{A_{r+1}}}.$$

By the recursive definition,

$$B_{r+1} = 3B_r + 2^{A_r}.$$

Therefore

$$T^{r+1}(n) = \frac{3^{r+1} n + B_{r+1}}{2^{A_{r+1}}}.$$

The induction is complete. □

Lemma 7.4 (Cumulative Descent Criterion)**.** *Let*

$$n \in \mathbb{O}^+$$

and suppose that the first r accelerated valuations of n are described by a pattern

$$\mathbf{a} = (a_0, \ldots, a_{r-1}).$$

Set

$$A = A_r(\mathbf{a}), \qquad B = B_r(\mathbf{a}).$$

Then

$$T^r(n) < n$$

if and only if

$$B < n(2^A - 3^r).$$

In particular, descent at time r requires

$$2^A > 3^r.$$

Proof. By Theorem 7.3,

$$T^r(n) = \frac{3^r n + B}{2^A}.$$

Thus

$$T^r(n) < n$$

is equivalent to

$$\frac{3^r n + B}{2^A} < n.$$

After multiplication by 2^A, one obtains

$$3^r n + B < 2^A n.$$

After rearrangement, this is equivalent to

$$B < n(2^A - 3^r).$$

This proves the equivalence.

Since

$$B \geq 0$$

and

$$n > 0,$$

the inequality

$$B < n(2^A - 3^r)$$

can hold only if

$$2^A - 3^r > 0.$$

Hence descent at time r requires

$$2^A > 3^r.$$

□

8 Residue Cylinders of Valuation Patterns

A finite valuation pattern imposes congruence conditions on the initial value. The next result shows that every valuation pattern determines a unique odd residue class modulo a power of 2. These residue classes organize the delayed-descent problem into congruence cylinders.

Definition 8.1 (Valuation Cylinder)**.** Let

$$\mathbf{a} = (a_0, \dots, a_{r-1})$$

be a valuation pattern of length $r \geq 1$. The valuation cylinder associated to $\mathbf{a}$ is

$$C(\mathbf{a}) = \left\{ n \in \mathbb{O}^+ : v_2(3T^i(n) + 1) = a_i \text{ for } 0 \leq i \leq r - 1 \right\}.$$

Theorem 8.2 (Residue Cylinder of a Valuation Pattern)**.** *Let*

$$\mathbf{a} = (a_0, \dots, a_{r-1})$$

be a valuation pattern of length $r \geq 1$, and set

$$A = A_r(\mathbf{a}).$$

Then there exists a unique odd residue class

$$\rho(\mathbf{a}) \pmod{2^{A+1}}$$

such that

$$n \in C(\mathbf{a})$$

if and only if

$$n \equiv \rho(\mathbf{a}) \pmod{2^{A+1}}.$$

Proof. The proof is by induction on the length r.

For $r = 1$, the condition is

$$v_2(3n + 1) = a_0.$$

Equivalently,

$$3n + 1 \equiv 2^{a_0} \pmod{2^{a_0+1}}.$$

Thus

$$3n \equiv 2^{a_0} - 1 \pmod{2^{a_0+1}}.$$

Since 3 is invertible modulo 2^{a_0+1}, this determines a unique residue class modulo 2^{a_0+1}. The residue is odd because $2^{a_0} - 1$ is odd.

Assume now that the statement holds for all patterns of length r. Let

$$\mathbf{a}' = (a_0, \dots, a_{r-1}, b)$$

be a pattern of length $r + 1$. Write

$$\mathbf{a} = (a_0, \dots, a_{r-1}), \qquad A = A_r(\mathbf{a}), \qquad A' = A + b.$$

By the induction hypothesis, the first r valuation conditions determine a unique odd residue class

$$\rho(\mathbf{a}) \pmod{2^{A+1}}.$$

Choose a representative $n_0 \in C(\mathbf{a})$. Then every lift of this residue class has the form

$$n = n_0 + 2^{A+1}t$$

for some integer t.

By the cumulative affine formula,

$$T^r(n) = \frac{3^r n + B_r(\mathbf{a})}{2^A}.$$

Therefore

$$T^r(n) = T^r(n_0) + 2 \cdot 3^r t.$$

The additional valuation condition is

$$v_2(3T^r(n) + 1) = b.$$

Equivalently,

$$3T^r(n) + 1 \equiv 2^b \pmod{2^{b+1}}.$$

Substituting the lifted form gives

$$3T^r(n_0) + 1 + 2 \cdot 3^{r+1}t \equiv 2^b \pmod{2^{b+1}}.$$

Both sides are even, so this is equivalent to the congruence modulo 2^b

$$\frac{3T^r(n_0) + 1}{2} + 3^{r+1}t \equiv 2^{b-1} \pmod{2^b}.$$

Since 3^{r+1} is odd, it is invertible modulo 2^b. Hence a unique value of

$$t \pmod{2^b}$$

is determined.

Thus the final valuation condition selects exactly one lift of

$$\rho(\mathbf{a}) \pmod{2^{A+1}}$$

to a residue class modulo

$$2^{A+1}2^b = 2^{A+b+1} = 2^{A'+1}.$$

This gives a unique odd residue class

$$\rho(\mathbf{a}') \pmod{2^{A'+1}}.$$

The induction is complete. □

Definition 8.3 (Non-Certified Valuation Cylinder)**.** Let

$$m \geq 2$$

and let $\mathbf{a}$ be a valuation pattern. The non-certified valuation cylinder at shell level m is

$$C_m(\mathbf{a}) = C(\mathbf{a}) \cap \mathcal{N}_m^{\circ}.$$

Lemma 8.4 (Congruence Form of Non-Certified Cylinders)**.** *Let*

$$m \geq 2$$

and let $\mathbf{a}$ *be a valuation pattern of length* r*, with*

$$A = A_r(\mathbf{a}).$$

Then

$$n \in C_m(\mathbf{a})$$

if and only if the following conditions hold:

$$n \equiv \rho(\mathbf{a}) \pmod{2^{A+1}},$$
$$n \equiv 2^m - 1 \pmod{2^{m+1}},$$

and

$$n \not\equiv 2^m q_m^* - 1 \pmod{2^{m+\kappa_m}}.$$

Proof. The first congruence is exactly the valuation-cylinder condition from Theorem 8.2.

Next,

$$n \in S_m^\circ$$

if and only if

$$v_2(n+1) = m.$$

This is equivalent to

$$n + 1 \equiv 2^m \pmod{2^{m+1}},$$

or

$$n \equiv 2^m - 1 \pmod{2^{m+1}}.$$

Finally, the certified component is described by

$$q \equiv q_m^* \pmod{2^{\kappa_m}}, \qquad n = 2^m q - 1.$$

Since

$$q = \frac{n+1}{2^m},$$

this congruence is equivalent to

$$n + 1 \equiv 2^m q_m^* \pmod{2^{m+\kappa_m}},$$

or

$$n \equiv 2^m q_m^* - 1 \pmod{2^{m+\kappa_m}}.$$

Thus the residual component $\mathcal{N}_m^\circ$ is obtained by excluding this certified residue class from the canonical shell.

Combining the valuation-cylinder congruence, the canonical-shell congruence, and the certified-residue exclusion gives the claim. □

Definition 8.5 (Cylinder Minimum)**.** If

$$C_m(\mathbf{a}) \neq \varnothing,$$

then its cylinder minimum is

$$n_{\min}(C_m(\mathbf{a})) = \min C_m(\mathbf{a}).$$

Remark 8.6. By Theorem 8.4, every nonempty non-certified valuation cylinder is described by congruence conditions modulo powers of 2, together with one excluded certified residue class. Hence

$$n_{\min}(C_m(\mathbf{a}))$$

is explicitly computable whenever the cylinder is nonempty.

9 Closed Cylinders and Uniform Descent

The cumulative descent criterion reduces descent at time r to

$$B_r(\mathbf{a}) < n\left(2^{A_r(\mathbf{a})} - 3^r\right).$$

For a fixed valuation cylinder, the quantities

$$A_r(\mathbf{a}) \qquad \text{and} \qquad B_r(\mathbf{a})$$

are constant. Hence, whenever

$$2^{A_r(\mathbf{a})} > 3^r,$$

the right-hand side

$$n\left(2^{A_r(\mathbf{a})} - 3^r\right)$$

is increasing as a function of n. Thus a descent certificate at the smallest element of the cylinder extends uniformly to the full cylinder.

Definition 9.1 (Closed Non-Certified Cylinder)**.** Let

$$C_m(\mathbf{a})$$

be a nonempty non-certified valuation cylinder of length r. Set

$$A = A_r(\mathbf{a}), \qquad B = B_r(\mathbf{a}), \qquad n_0 = n_{\min}(C_m(\mathbf{a})).$$

The cylinder

$$C_m(\mathbf{a})$$

is called closed if

$$2^A > 3^r$$

and

$$B < n_0(2^A - 3^r).$$

Theorem 9.2 (Closed Cylinder Gives Uniform Descent)**.** *Let*

$$C_m(\mathbf{a})$$

be a closed non-certified valuation cylinder of length r. Then every

$$n \in C_m(\mathbf{a})$$

satisfies

$$T^r(n) < n.$$

Proof. Let

$$n \in C_m(\mathbf{a}).$$

Set

$$A = A_r(\mathbf{a}), \qquad B = B_r(\mathbf{a}), \qquad n_0 = n_{\min}(C_m(\mathbf{a})).$$

Since

$$n_0$$

is the minimum element of the cylinder,

$$n \geq n_0.$$

Closedness gives

$$2^A > 3^r,$$

so

$$2^A - 3^r > 0.$$

Therefore

$$n(2^A - 3^r) \geq n_0(2^A - 3^r).$$

By the defining descent inequality for a closed cylinder,

$$B < n_0(2^A - 3^r).$$

Combining the two inequalities gives

$$B < n(2^A - 3^r).$$

Since every element of

$$C_m(\mathbf{a})$$

has valuation pattern

$$\mathbf{a}$$

for the first r accelerated steps, Theorem 7.4 gives

$$T^r(n) < n.$$

□

Corollary 9.3 (Closed Non-Certified Cylinders Give Delayed Descent)**.** *Assume the dyadic gap condition at level m. Let*

$$C_m(\mathbf{a})$$

be a closed non-certified valuation cylinder of length r. Then every

$$n \in C_m(\mathbf{a})$$

satisfies

$$T^r(n) < n.$$

Moreover, this descent is delayed:

$$r > m.$$

Proof. By Theorem 9.2,

$$T^r(n) < n$$

for every

$$n \in C_m(\mathbf{a}).$$

Since

$$C_m(\mathbf{a}) \subseteq \mathcal{N}_m^\circ,$$

Theorem 5.7 gives

$$r > m.$$

□

Proposition 9.4 (Closed Cylinder Cover Gives Non-Certified Descent)**.** *Fix*

$$m \geq 2.$$

Suppose that the non-certified complement is covered by closed valuation cylinders:

$$\mathcal{N}_m^\circ \subseteq \bigcup_{\alpha \in I} C_m(\mathbf{a}_\alpha),$$

where each

$$C_m(\mathbf{a}_\alpha)$$

is closed. Then

$$\forall n \in \mathcal{N}_m^\circ, \qquad \exists r \geq 1 : T^r(n) < n.$$

If the dyadic gap condition holds at level m, then the descent is delayed:

$$\exists r > m : T^r(n) < n.$$

Proof. Let

$$n \in \mathcal{N}_m^\circ.$$

By the covering assumption, there exists

$$\alpha \in I$$

such that

$$n \in C_m(\mathbf{a}_\alpha).$$

Let

$$r_\alpha$$

be the length of

$$\mathbf{a}_\alpha.$$

Since

$$C_m(\mathbf{a}_\alpha)$$

is closed, Theorem 9.2 gives

$$T^{r_\alpha}(n) < n.$$

Thus n has finite descent.

If the dyadic gap condition holds at level m, then Theorem 5.7 gives

$$r_\alpha > m.$$

□

Remark 9.5 (Uniform Nature of Closed Cylinders)**.** A closed cylinder gives a uniform descent certificate. Indeed, all elements of

$$C_m(\mathbf{a})$$

share the same cumulative data

$$A_r(\mathbf{a}), \qquad B_r(\mathbf{a}).$$

When

$$2^{A_r(\mathbf{a})} > 3^r,$$

the quantity

$$n(2^{A_r(\mathbf{a})} - 3^r)$$

is increasing in n. Therefore verifying the descent inequality at

$$n_{\min}(C_m(\mathbf{a}))$$

certifies descent for the entire cylinder.

10 Non-Certified Residue-Cover Trees

Non-certified valuation cylinders are organized into a recursive residue-cover tree. Each active cylinder is refined by appending one additional valuation. When every branch reaches a closed cylinder, each non-certified element receives a cumulative descent certificate.

Definition 10.1 (Non-Certified Residue-Cover Tree)**.** Fix

$$m \geq 2.$$

The non-certified residue-cover tree

$$\mathcal{T}_m$$

is defined as follows.

The root is the full non-certified canonical-shell component

$$\mathcal{N}_m^{\circ}.$$

A node at depth $r \geq 1$ is a nonempty valuation cylinder

$$C_m(\mathbf{a}),$$

where

$$\mathbf{a} = (a_0, \ldots, a_{r-1})$$

is a valuation pattern of length r.

A node is called closed if it is closed in the sense of

$$Theorem\ 9.1.$$

A node that is still to be refined is called active.

Definition 10.2 (Children of an Active Node)**.** Let

$$C_m(\mathbf{a})$$

be an active node of depth r, where

$$\mathbf{a} = (a_0, \ldots, a_{r-1}).$$

For each integer

$$b \geq 1,$$

write

$$\mathbf{a} * b = (a_0, \ldots, a_{r-1}, b).$$

The children of

$$C_m(\mathbf{a})$$

are the nonempty cylinders

$$C_m(\mathbf{a} * b).$$

Proposition 10.3 (One-Step Cylinder Refinement)**.** *Let*

$$C_m(\mathbf{a})$$

be a nonempty non-certified valuation cylinder of depth r. Then

$$C_m(\mathbf{a}) = \bigsqcup_{b \geq 1} C_m(\mathbf{a} * b),$$

where empty cylinders are omitted.

Proof. Let

$$n \in C_m(\mathbf{a}).$$

Then the first r accelerated valuations of n are fixed by

$$\mathbf{a} = (a_0, \ldots, a_{r-1}).$$

The next valuation is

$$b = v_2(3T^r(n) + 1).$$

Since

$$T^r(n)$$

is odd, the integer

$$3T^r(n) + 1$$

is even. Hence

$$b \geq 1.$$

Therefore

$$n \in C_m(\mathbf{a} * b).$$

Conversely, each cylinder

$$C_m(\mathbf{a} * b)$$

is obtained from

$$C_m(\mathbf{a})$$

by imposing one additional valuation condition, so

$$C_m(\mathbf{a} * b) \subseteq C_m(\mathbf{a}).$$

The union is disjoint because the next valuation

$$v_2(3T^r(n) + 1)$$

is uniquely determined for each

$$n \in C_m(\mathbf{a}).$$

Thus the nonempty children form a disjoint refinement of the parent cylinder. □

Definition 10.4 (Branch Generated by a Non-Certified Element)**.** Let

$$n \in \mathcal{N}_m^\circ.$$

The branch of

$$\mathcal{T}_m$$

generated by n is the sequence of valuation patterns

$$\mathbf{a}^{(r)}(n) = (a_0(n), \ldots, a_{r-1}(n)), \qquad r \geq 1,$$

where

$$a_i(n) = v_2(3T^i(n) + 1).$$

Equivalently, the depth-r node on this branch is

$$C_m(\mathbf{a}^{(r)}(n)).$$

Definition 10.5 (Tree Closure)**.** The non-certified residue-cover tree

$$\mathcal{T}_m$$

is said to close if every branch generated by an element

$$n \in \mathcal{N}_m^\circ$$

contains a closed node.

Equivalently, for every

$$n \in \mathcal{N}_m^\circ,$$

there exists an integer

$$r \geq 1$$

such that

$$C_m(\mathbf{a}^{(r)}(n))$$

is closed.

Theorem 10.6 (Tree Closure Implies Non-Certified Descent)**.** *Fix*

$$m \geq 2.$$

If the non-certified residue-cover tree

$$\mathcal{T}_m$$

closes, then

$$\forall n \in \mathcal{N}_m^\circ, \qquad \exists r \geq 1 : T^r(n) < n.$$

Proof. Let

$$n \in \mathcal{N}_m^\circ.$$

Since

$$\mathcal{T}_m$$

closes, the branch generated by n contains a closed node. Thus there exists

$$r \geq 1$$

such that

$$C_m(\mathbf{a}^{(r)}(n))$$

is closed. By construction,

$$n \in C_m(\mathbf{a}^{(r)}(n)).$$

Applying Theorem 9.2 gives

$$T^r(n) < n.$$

□

Theorem 10.7 (Tree Closure Implies Conditional Non-Certified Delayed Coverage)**.** *Fix*

$$m \geq 2.$$

Assume the dyadic gap condition at level m. If the non-certified residue-cover tree

$$\mathcal{T}_m$$

closes, then

$$\forall n \in \mathcal{N}_m^\circ, \qquad \exists r > m : T^r(n) < n.$$

Proof. Let

$$n \in \mathcal{N}_m^\circ.$$

By Theorem 10.6, there exists

$$r \geq 1$$

such that

$$T^r(n) < n.$$

Since the dyadic gap condition holds at level m, Theorem 5.7 gives

$$r > m.$$

Therefore

$$\exists r > m : T^r(n) < n.$$

□

Proposition 10.8 (Closed Nodes Form a Coverage Certificate)**.** *Fix*

$$m \geq 2.$$

Suppose that the non-certified component is covered by closed cylinders:

$$\mathcal{N}_m^\circ \subseteq \bigcup_{\alpha \in I} C_m(\mathbf{a}_\alpha),$$

where each

$$C_m(\mathbf{a}_\alpha)$$

is a closed node of

$$\mathcal{T}_m.$$

Then

$$\mathcal{T}_m$$

closes.

Proof. Let

$$n \in \mathcal{N}_m^\circ.$$

By the covering assumption, there exists

$$\alpha \in I$$

such that

$$n \in C_m(\mathbf{a}_\alpha).$$

Since

$$C_m(\mathbf{a}_\alpha)$$

is a node of

$$\mathcal{T}_m,$$

the valuation pattern

$$\mathbf{a}_\alpha$$

is an initial valuation pattern of n. Hence the branch generated by n contains the node

$$C_m(\mathbf{a}_\alpha).$$

This node is closed by assumption. Therefore every branch generated by an element of

$$\mathcal{N}_m^\circ$$

contains a closed node, and so

$$\mathcal{T}_m$$

closes. □

Remark 10.9 (Role of Tree Closure)**.** Tree closure is the structural mechanism that converts local cylinder certificates into full non-certified delayed coverage. A closed cylinder gives a uniform descent certificate for every element in that cylinder. Therefore, when every branch of

$$\mathcal{T}_m$$

reaches a closed cylinder, the entire non-certified component

$$\mathcal{N}_m^\circ$$

is covered by delayed-descent certificates.

11 Conditional Full Non-Certified Delayed Coverage

The preceding sections reduce delayed descent in the non-certified component to two structural ingredients. The dyadic gap condition gives the first-exit obstruction

$$T^m(n) > n \qquad (n \in \mathcal{N}_m^\circ),$$

so any descent in the non-certified component occurs strictly after the deterministic exit time m. Tree closure gives a closed valuation cylinder on every non-certified branch, and each closed cylinder provides a uniform cumulative descent certificate.

Combining these two ingredients gives conditional delayed-descent coverage of the full non-certified component.

Theorem 11.1 (Conditional Full Non-Certified Delayed Coverage)**.** *Assume the dyadic gap condition globally. Suppose that, for every*

$$m \geq 2,$$

the non-certified residue-cover tree

$$\mathcal{T}_m$$

closes. Then

$$\forall m \geq 2, \quad \forall n \in \mathcal{N}_m^\circ, \quad \exists r > m : T^r(n) < n.$$

Equivalently, every non-certified canonical-shell element has delayed descent.

Proof. Fix

$$m \geq 2$$

and let

$$n \in \mathcal{N}_m^\circ.$$

By the global dyadic gap condition, the dyadic gap holds at level m. By hypothesis, the tree

$$\mathcal{T}_m$$

closes. Therefore Theorem 10.7 gives an integer

$$r > m$$

such that

$$T^r(n) < n.$$

Since m and n were arbitrary, the conclusion holds for every

$$m \geq 2$$

and every

$$n \in \mathcal{N}_m^\circ.$$

□

Corollary 11.2 (Conditional Coverage from Closed Cylinder Covers)**.** *Assume the dyadic gap condition globally. Suppose that, for every*

$$m \geq 2,$$

the non-certified component admits a cover by closed valuation cylinders:

$$\mathcal{N}_m^\circ \subseteq \bigcup_{\alpha \in I_m} C_m(\mathbf{a}_\alpha),$$

where each

$$C_m(\mathbf{a}_\alpha)$$

is a closed node of

$$\mathcal{T}_m.$$

Then

$$\forall m \geq 2, \quad \forall n \in \mathcal{N}_m^\circ, \quad \exists r > m : T^r(n) < n.$$

Proof. Fix
$$m \geq 2.$$
By the closed-cylinder cover and Theorem 10.8, the tree
$$\mathcal{T}_m$$
closes. Since this holds at every level
$$m \geq 2,$$
Theorem 11.1 gives
$$\forall m \geq 2, \quad \forall n \in \mathcal{N}_m^{\circ}, \quad \exists r > m : T^r(n) < n.$$
□

Corollary 11.3 (Pointwise Non-Certified Coverage Criterion)**.** *Assume the dyadic gap condition globally. Suppose that, for every*
$$m \geq 2$$
and every
$$n \in \mathcal{N}_m^{\circ},$$
there exists an integer $r \geq 1$ *such that the finite valuation pattern*
$$\mathbf{a}^{(r)}(n) = (a_0(n), \dots, a_{r-1}(n))$$
determines a closed cylinder
$$C_m(\mathbf{a}^{(r)}(n)).$$
Then
$$\forall m \geq 2, \quad \forall n \in \mathcal{N}_m^{\circ}, \quad \exists r > m : T^r(n) < n.$$
Proof. The hypothesis states that every branch generated by an element of
$$\mathcal{N}_m^{\circ}$$
contains a closed node. Hence every tree
$$\mathcal{T}_m$$
closes. The conclusion follows from Theorem 11.1. □

Remark 11.4 (Status of the Conditional Coverage Theorem)**.** The theorem identifies two structural ingredients for full non-certified delayed-descent coverage:

the global dyadic gap condition

and

closure of every non-certified residue-cover tree.

Together, these ingredients give
$$\forall m \geq 2, \quad \forall n \in \mathcal{N}_m^{\circ}, \quad \exists r > m : T^r(n) < n.$$
Thus the non-certified delayed-descent problem is reduced to dyadic-gap control and residue-cover tree closure.

12 Conditional Universal First-Descent Coverage

The canonical shell decomposition separates the positive odd integers into disjoint regimes:

$$\mathbb{O}^{+} = S_1^{\circ} \sqcup \bigsqcup_{m \geq 2} S_m^{\circ}.$$

The first shell gives immediate descent. For each nontrivial shell

$$S_m^{\circ} \qquad (m \geq 2),$$

the certified/non-certified decomposition is

$$S_m^{\circ} = \mathcal{D}_m^{\circ} \sqcup \mathcal{N}_m^{\circ}.$$

Certified elements descend exactly at the deterministic exit time m. Under the dyadic gap condition and residue-cover tree closure, non-certified elements have delayed descent.

Lemma 12.1 (Immediate Descent in the First Canonical Shell)**.** *Let*

$$n \in S_1^{\circ}$$

with

$$n > 1.$$

Then

$$T(n) < n.$$

Consequently,

$$\tau(n) = 1.$$

Proof. Since

$$n \in S_1^{\circ},$$

one has

$$v_2(n+1) = 1.$$

Equivalently,

$$n \equiv 1 \pmod{4}.$$

Therefore

$$3n + 1 \equiv 3 \cdot 1 + 1 \equiv 0 \pmod{4},$$

so

$$v_2(3n+1) \geq 2.$$

Hence

$$T(n) = \frac{3n+1}{2^{v_2(3n+1)}} \leq \frac{3n+1}{4}.$$

Since

$$n > 1,$$

one has

$$\frac{3n+1}{4} < n.$$

Thus

$$T(n) < n.$$

Since this descent occurs at the first accelerated step,

$$\tau(n) = 1.$$

□

Theorem 12.2 (Conditional Universal First-Descent Coverage)**.** *Assume the dyadic gap condition globally. Suppose that, for every*

$$m \geq 2,$$

the non-certified residue-cover tree

$$\mathcal{T}_m$$

closes. Then every positive odd integer

$$n > 1$$

has a descent below its initial value:

$$\forall n \in \mathbb{O}^+,\ n > 1, \qquad \exists r \geq 1 : T^r(n) < n.$$

Proof. Let

$$n \in \mathbb{O}^+, \qquad n > 1.$$

By Theorem 2.3, there is a unique

$$m \geq 1$$

such that

$$n \in S_m^\circ.$$

If

$$m = 1,$$

then Theorem 12.1 gives

$$T(n) < n.$$

Thus the conclusion holds with

$$r = 1.$$

Now let

$$m \geq 2.$$

By Theorem 4.4,

$$S_m^\circ = \mathcal{D}_m^\circ \sqcup \mathcal{N}_m^\circ.$$

If

$$n \in \mathcal{D}_m^\circ,$$

then Theorem 4.5 gives

$$\tau(n) = m,$$

and hence

$$T^m(n) < n.$$

If

$$n \in \mathcal{N}_m^\circ,$$

then Theorem 11.1 gives an integer

$$r > m$$

such that

$$T^r(n) < n.$$

Thus every canonical-shell case gives a finite first descent below the initial value. $\square$

Corollary 12.3 (Conditional First-Descent Partition)**.** *Assume the dyadic gap condition globally and suppose that*

$$\mathcal{T}_m$$

closes for every

$$m \geq 2.$$

Then every positive odd integer

$$n > 1$$

belongs to exactly one of the following descent regimes:

$$n \in S_1^\circ \quad \Longrightarrow \quad \tau(n) = 1,$$

$$n \in \mathcal{D}_m^\circ \quad \Longrightarrow \quad \tau(n) = m,$$

and

$$n \in \mathcal{N}_m^\circ \quad \Longrightarrow \quad \tau(n) > m.$$

In all cases,

$$\tau(n) < \infty.$$

Proof. The first regime follows from Theorem 12.1. The second follows from Theorem 4.5. For the third regime, Theorem 11.1 gives a descent time

$$r > m$$

for every

$$n \in \mathcal{N}_m^\circ,$$

while Theorem 5.7 shows that every descent time in the non-certified component is strictly larger than

$$m.$$

Hence

$$\tau(n) > m$$

for

$$n \in \mathcal{N}_m^\circ.$$

The regimes are disjoint and exhaustive by Theorem 2.3 and Theorem 4.4. $\square$

13 Relation to the Collatz Conjecture

The previous section showed that the global dyadic gap condition, together with closure of every non-certified residue-cover tree, implies universal first-descent coverage for the accelerated odd Collatz map. The standard minimal-counterexample argument then converts universal first descent into convergence to 1 [4, 8].

Recall that the accelerated odd Collatz map is

$$T(n) = \frac{3n+1}{2^{\nu_2(3n+1)}},$$

defined on the positive odd integers. The usual Collatz iteration reaches 1 for every positive integer if and only if every positive odd integer reaches 1 under iteration of T.

Theorem 13.1 (Universal First-Descent Coverage Implies Collatz Convergence)**.** *Suppose that every positive odd integer*

$$n > 1$$

has a finite descent under the accelerated odd map:

$$\forall n \in \mathbb{O}^+,\ n > 1, \qquad \exists r \geq 1 : T^r(n) < n.$$

Then every positive odd integer eventually reaches 1 *under iteration of* T*. Consequently, every positive integer eventually reaches* 1 *under the usual Collatz iteration.*

Proof. Assume that there is a positive odd integer whose forward orbit under

$$T$$

avoids 1. Let

$$C = \{n \in \mathbb{O}^+ : n > 1 \text{ and the forward orbit of } n \text{ under } T \text{ avoids } 1\}.$$

Then

$$C$$

is a nonempty subset of the positive integers, so it has a least element. Write

$$n_0 = \min C.$$

By universal first-descent coverage, there exists

$$r \geq 1$$

such that

$$T^r(n_0) < n_0.$$

Set

$$n_1 = T^r(n_0).$$

Then

$$n_1$$

is a positive odd integer with

$$n_1 < n_0.$$

Since

$$n_1$$

lies on the forward orbit of

$$n_0,$$

convergence of the orbit of n_1 to 1 would imply convergence of the orbit of n_0 to 1. Thus

$$n_1 \in C.$$

But this contradicts the minimality of

$$n_0 = \min C,$$

because

$$n_1 < n_0.$$

Therefore

$$C = \varnothing.$$

Hence every positive odd integer eventually reaches 1 under iteration of

$$T.$$

Finally, every positive integer reaches a positive odd integer after finitely many divisions by 2. Since every accelerated odd trajectory reaches 1, the usual Collatz trajectory of every positive integer also reaches 1. □

Corollary 13.2 (Conditional Route to Collatz Convergence)**.** *Assume the dyadic gap condition globally. Suppose that, for every*

$$m \geq 2,$$

the non-certified residue-cover tree

$$\mathcal{T}_m$$

closes. Then every positive integer eventually reaches 1 *under the usual Collatz iteration.*

Proof. By Theorem 12.2, the global dyadic gap condition and closure of every non-certified residue-cover tree imply universal first-descent coverage:

$$\forall n \in \mathbb{O}^+,\ n > 1, \qquad \exists r \geq 1 : T^r(n) < n.$$

By Theorem 13.1, universal first-descent coverage implies convergence to 1 for every positive odd integer under the accelerated odd map. Consequently, every positive integer converges to 1 under the usual Collatz iteration. □

Remark 13.3 (Conditional Route)**.** The preceding corollary records the conditional implication

$$\text{Global Dyadic Gap} + \text{Non-Certified Tree Closure} \Longrightarrow \text{Universal First Descent} \Longrightarrow \text{Collatz Convergence}.$$

Thus the framework reduces the convergence route to two structural tasks:

proving the global dyadic gap condition

and

proving closure of every non-certified residue-cover tree.

14 Computational Evidence

The preceding sections reduce full non-certified delayed coverage to two structural components: the dyadic gap condition and closure of the non-certified residue-cover trees. This section reports finite-range exact computations supporting both components.

Two computations were performed. First, the dyadic gap condition was verified over a finite range of shell levels and compared with the continued-fraction obstruction described in Section 6. Second, canonical non-certified shell elements were tested for delayed first descent up to a finite search bound.

14.1 Finite Verification of the Dyadic Gap

The dyadic gap condition is

$$2^{K_m} - 3^m \geq 2^{K_m - m}, \qquad K_m = \lceil m \log_2 3 \rceil .$$

The finite verification was performed for

$$2 \leq m \leq 100000.$$

The computation used 800 decimal digits of precision to propose values of

$$K_m = \lceil m \log_2 3 \rceil .$$

Each proposed value was independently validated by the exact integer inequalities

$$2^{K_m - 1} < 3^m < 2^{K_m}.$$

After this validation, the dyadic margin

$$D_m = 2^{K_m} - 3^m - 2^{K_m - m}$$

was computed using exact integer arithmetic.

The direct dyadic gap test checked

$$99999$$

levels. In this range,

$$D_m \geq 0 \qquad (2 \leq m \leq 100000),$$

and every proposed value of

$$K_m$$

passed the exact integer validation.

The continued-fraction obstruction identified in Theorem 6.6 was also checked. The computation used 120 continued-fraction terms for

$$\log_2 3$$

and tested the upper convergent denominators satisfying

$$2 \leq q \leq 100000.$$

The upper convergent denominators in this range were

$$5, \quad 41, \quad 306, \quad 15601, \quad 79335.$$

At these denominators, the corresponding values of

$$K_m$$

were

$$8, \quad 65, \quad 485, \quad 24727, \quad 125743.$$

Each value passed the exact integer validation, and the dyadic margin was positive at every tested obstruction level.

Quantity	Value
Precision	800 decimal digits
Direct dyadic gap range	$2 \le m \le 100000$
Checked levels	99999
Dyadic gap failures	0
Invalid K_m values	0
Continued-fraction terms used	120
Upper convergent denominator bound	$2 \le q \le 100000$
Upper convergent denominators checked	$5, 41, 306, 15601, 79335$
Candidate failures at upper convergents	0

Table 1: Finite verification of the dyadic gap condition and its continued-fraction obstruction.

These computations provide finite-range support for the dyadic gap condition. They verify the condition for

$$2 \le m \le 100000,$$

and, within the continued-fraction reduction of Theorem 6.6, they also verify all upper convergent obstruction levels in this range.

14.2 Canonical Non-Certified Shell Computation

The canonical-shell delayed-descent computation was performed up to

$$N = 10^7$$

with accelerated-iteration limit

$$L = 10000.$$

The admissible nontrivial canonical shell levels in this range are

$$2 \le m \le 23.$$

Indeed,

$$2^{23} - 1 = 8,388,607 < 10^7, \qquad 2^{24} - 1 = 16,777,215 > 10^7.$$

Unlike nested spine-level enumeration, canonical-shell enumeration assigns each positive odd integer to a unique shell. Thus each tested odd integer is counted at most once. The tested nontrivial canonical-shell population is

$$\sum_{m=2}^{23} |S_m^{\circ}(10^7)| = 2,500,000.$$

This population decomposes as

$$2,500,000 = 1,068,631 + 1,431,369,$$

where

$$|\mathcal{D}^{\circ}(10^7)| = 1,068,631$$

and

$$|\mathcal{N}^{\circ}(10^7)| = 1,431,369.$$

For every tested canonical non-certified shell element

$$n \in \mathcal{N}_m^{\circ},$$

the first-descent time

$$\tau(n) = \min\{r \geq 1 : T^r(n) < n\}$$

was computed using exact integer arithmetic. The delayed excess was recorded as

$$E(n) = \tau(n) - m.$$

Quantity	Value
Search bound N	$10,000,000$
Step limit L	$10,000$
Admissible shell levels	$2 \leq m \leq 23$
Canonical shell cases with $m \geq 2$	$2,500,000$
Certified canonical-shell cases	$1,068,631$
Canonical non-certified shell cases	$1,431,369$
Resolved non-certified cases	$1,431,369$
Unresolved cases	0
Early descents $\tau < m$	0
Exact descents $\tau = m$	0
Delayed descents $\tau > m$	$1,431,369$
Cumulative descent-check failures at τ	0

Table 2: Exact-integer verification summary for canonical non-certified shell elements up to $N = 10^7$.

Within the tested canonical-shell range, every evaluated non-certified shell element was resolved within the iteration limit. Each resolved non-certified case exhibited delayed descent:

$$\tau(n) > m.$$

Thus the tested data are consistent with the first-exit obstruction developed above: non-certified shell elements pass the deterministic exit time before descending.

For every resolved case, the cumulative descent inequality held at the computed first-descent time:

$$B_{\tau(n)}(n) < n\left(2^{A_{\tau(n)}(n)} - 3^{\tau(n)}\right).$$

This provides an implementation consistency check for the cumulative affine formula at the computed first-descent time.

Statistic	Value
Mean $\tau(n)$	8.4994568137
Median $\tau(n)$	6
Maximum $\tau(n)$	155
Mean $E(n) = \tau(n) - m$	5.2058944968
Median $E(n)$	2
Maximum $E(n)$	146

Table 3: First-descent and delayed-excess statistics for canonical non-certified shell elements up to $N = 10^7$.

The per-shell count decomposition and per-shell delayed-descent statistics are included in the accompanying computational output files. The per-shell data show increasing average delay across the populated shell levels, while the number of tested elements decreases rapidly as m increases. Accordingly, the high-m rows are best interpreted as endpoint-limited finite-range observations.

These computations provide finite-range evidence for delayed descent in the canonical non-certified component. They support the structural picture developed above: certified shell elements descend exactly at the deterministic exit time, while tested non-certified shell elements pass the deterministic exit and descend later through cumulative valuation.

15 Conclusion

This paper developed a conditional residue-tree framework for delayed descent in the accelerated odd Collatz map. The positive odd integers were decomposed into canonical shells

$$S_m^\circ = \{n \in \mathbb{O}^+ : v_2(n+1) = m\}.$$

Within each nontrivial shell

$$m \geq 2,$$

deterministic persistence shows that descent below the initial value occurs only at or after the deterministic exit time

$$m.$$

At the exit time, each nontrivial shell was decomposed into certified and non-certified components:

$$S_m^\circ = \mathcal{D}_m^\circ \sqcup \mathcal{N}_m^\circ.$$

Certified elements descend exactly at time

$$m,$$

while non-certified elements, under the dyadic gap condition, remain above their initial value at the deterministic exit time. Hence descent in the non-certified component is delayed beyond the exit.

The dyadic gap condition was reduced to a sparse Diophantine obstruction. A dyadic-gap failure forces an upper continued-fraction convergent approximation to

$$\log_2 3.$$

Thus dyadic-gap obstruction levels are constrained to a sparse continued-fraction set.

The delayed-descent problem in the non-certified component was then formulated through valuation cylinders and non-certified residue-cover trees. A closed cylinder provides a uniform cumulative descent certificate for every element in that cylinder. Consequently, closure of every non-certified residue-cover tree gives delayed descent throughout the non-certified complement.

The central conditional route established in this paper is

$$\text{Global Dyadic Gap} + \text{Non-Certified Tree Closure} \implies \text{Full Non-Certified Delayed Coverage}.$$

Combining this with certified exact descent in the certified shell components and immediate descent in the first canonical shell gives

$$\text{Global Dyadic Gap} + \text{Non-Certified Tree Closure} \implies \text{Universal First Descent}.$$

Finally, by the standard minimal-counterexample argument,

$$\text{Universal First Descent} \implies \text{Collatz Convergence}.$$

Thus the full conditional implication obtained by the framework is

$$\text{Global Dyadic Gap} + \text{Non-Certified Tree Closure} \implies \text{Collatz Convergence}.$$

Accordingly, the framework isolates two structural requirements:

$$\text{the global dyadic gap condition}$$

and

$$\text{closure of every non-certified residue-cover tree.}$$

Establishing these two requirements would complete the route from the present framework to Collatz convergence.

Finite-range computations support the proposed structure. The dyadic gap condition was verified for

$$2 \leq m \leq 100000.$$

In the canonical non-certified shell computation up to

$$N = 10^7,$$

all

$$1{,}431{,}369$$

tested non-certified cases were resolved within the iteration limit. Each resolved non-certified case exhibited delayed descent:

$$\tau(n) > m.$$

These computations are consistent with the structural picture developed in the paper: certified elements descend at the deterministic exit, while tested non-certified elements pass the exit and descend later through cumulative valuation.

The framework developed here does not prove the Collatz conjecture unconditionally. Rather, it reduces the first-descent problem to two explicit structural tasks: the global dyadic gap condition and closure of the non-certified residue-cover trees. The finite computations reported in this paper support the proposed mechanism over the tested range, but they remain finite evidence only. A full proof through this route would require establishing both structural conditions for all shell levels and all non-certified branches.

Declarations

Data and code availability. The supporting code associated with this work is available in the public GitHub repository

`https://github.com/seymus1/conditional-collatz-shells`.

A permanent archival record will be deposited on Zenodo after release. The repository contains the script used for the canonical non-certified shell delayed-descent computation.

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